

Avikshith Yelakonda

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Professional Summary

Data professional with 3 years of experience transforming complex datasets into actionable insights that drive measurable business impact. Proficient in Python, SQL, visualization tools, machine learning, big data platforms, AI techniques, and ETL/data warehousing workflows. Skilled in designing data models, data preprocessing, and applying ML/DL & statistical methods to optimize performance and decision-making. Recognized for a proactive, adaptable approach in commitment to solving real-world challenges.

Skills

Programming & Tools: Python (pandas, NumPy, scikit-learn), R, SAS, SQL (Snowflake, Databricks, SQL Server), Git, Bash

ETL & Data Engineering: Databricks, Airflow, dbt, SQL pipelines, API ingestion, Data Modeling (Star/Snowflake), Data Quality & Validation

Machine Learning & Analytics: Predictive Modeling, Regression, Classification, Clustering, Time Series, Feature Engineering, HPO, Model Validation (AUC, RMSE, F1), Tuning

NLP & RAG: Text Mining, spaCy, Hugging Face, LangChain, OpenAI API, Retrieval-Augmented Generation, Prompt Engineering

Visualization & Reporting: Power BI (DAX, KPI Dashboards), Tableau, Excel (Power Query, Pivots), KPI Design, Data Storytelling

Cloud & Big Data: Azure (Synapse, Data Factory), AWS (S3, Glue), Snowflake, Spark, Docker, REST APIs, MLflow/Experiment Tracking

Statistics & Experimentation: EDA, Hypothesis Testing, ANOVA, Bayesian Inference, Forecasting, Uncertainty Quantification, A/B Testing

Governance & Compliance: AI Governance, Data Privacy & PII, Audit Trails, Documentation, SDLC Practices, Reproducibility

Experience

PioneerSoft (Client: Freddie Mac) | Business Data Scientist | USA

July 2025 – Present

- Designed ETL pipelines in Databricks & Snowflake to integrate structured & unstructured data for loan performance analytics.
- Built predictive models in Python & R using regression & tree-based algorithms to forecast credit risk & default likelihood.
- Developed automated data validation workflows & quality checks to ensure accuracy & compliance across financial datasets.
- Applied feature engineering & EDA to extract insights from servicing data, improving model performance by 21%.
- Created Power BI dashboards visualizing KPIs on asset quality, portfolio risk, & operational efficiency for stakeholders.
- Deployed MLflow-based model tracking pipelines for model validation & evaluation (AUC, RMSE, F1, Precision/Recall).
- Leveraged Azure Synapse & AWS S3 for scalable data storage, transformation, & orchestration across distributed systems.
- Collaborated with field data teams to automate ingestion of service metrics & improve end-to-end reporting efficiency.
- Applied predictive modeling and statistical analysis in Python & R to identify failure patterns and optimize service operations.

Yogin | Data Analyst | India

May 2021 – May 2023

- Built Power BI dashboards with advanced DAX for tracking inventory, warehouse throughput, and cost-to-serve metrics.
- Designed ETL pipelines in Python & Databricks to automate data ingestion from ERP, WMS, and timekeeping systems.
- Cleaned & modeled data in SQL Server for trend analysis and efficiency improvement across logistics operations.
- Applied statistical analysis & regression models to identify key performance drivers & reduce cost variance.
- Created workflow automations in Power Automate & Python to reduce manual reporting time by over 40%.
- Developed data validation scripts ensuring accuracy and consistency across HR, Finance, and Operations datasets.
- Collaborated with site leadership to implement data-backed recommendations that improved KPI achievement rates.
- Designed semantic data models & star schemas in Power BI for scalable, self-service business intelligence.
- Built NLP workflows to transform unstructured text from reports into structured insights using LangChain & regex parsing.

Projects

Consumer Affairs Lead Conversion Case Study | Python, Scikit-learn, XGBoost, Power BI

- Built an end-to-end **predictive modeling pipeline** (Logistic Regression, XGBoost) with **ROC AUC $\approx$ 0.68**, enabling **2.3–2.4 $\times$ lift in top-decile leads** and delivering actionable insights via **business-ready dashboards**.

Machine Learning & AI with Python Course Projects | Python, NumPy, TensorFlow, Keras, CNN, Transformers

- Built and optimized ML & DL models (Logistic Regression, MLPs, CNNs, Transformers) using NumPy, TensorFlow, and Keras. Applied PCA, feature extraction, adaptive learning, and regularization on diverse datasets, showcasing full pipeline design. Developed AI agents solving search & optimization problems (maze, n-Queens, Connect 4) using BFS, DFS, A*, Hill Climbing, and Minimax.

AI-Powered News Research Assistant | Python, LangChain, FAISS, OpenAI

- Reduced financial research time by 40% by building a **custom NLP assistant** that used **LangChain** and **FAISS** for **semantic search and document retrieval** in financial news and documents.

Revenue Growth Campaign Optimization via A/B Testing | Python, A/B Testing, KPI Analysis

- Increased marketing ROI by 22%** by designing and analyzing **A/B tests** across Google vs. Facebook ads, using statistical metrics to optimize budget allocation and conversion rates.

Stock Price Time-Series Forecasting | Python, ARIMA, Data Forecasting

- Predicted stock movements** for Shopify and Alibaba by applying **ARIMA and exponential smoothing models**, aiding **risk-based investor profiling** and investment decision support.

City Sanitation Multi-Source RAG Assistant | Python, LangChain, LlamaIndex, OpenAI, FAISS, Streamlit, RAG

- Built an **Advanced RAG system** integrating **web, PDF, and CSV data** using **LangChain**, **LlamaIndex**, and **FAISS** to deliver concise, citation-based answers on city sanitation and recycling services.

Education

Southern Methodist University | MS in Computer Science – AI/ML | GPA: 3.5/4.0 |